

MongoDB Lab Manual



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai



**Department of Information
Technology**

Laboratory Record Regulation - 2023

MongoDB Essentials

IT23431



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DEPARTMENT OF INFORMATION TECHNOLOGY

IT23431 - MONGODB ESSENTIALS

(Regulation 2023)

LAB MANUAL

Name :

Register No. :

Year / Branch / Section :

Semester :

Academic Year :

Course Code	Course Title (Laboratory Integrated Theory Course)	L	T	P	C
IT23431	Mongo DB Essentials	2	0	2	3

LIST OF EXPERIMENTS			
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2	Creating and dropping tables with various constraints		
3	Performing CRUD Operations		
4	Applying CRUD Operations		
5	Applying filtering and projections		
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7	Building aggregation pipelines		
8	Performing Data Transformation		
9	Establishing connection between Mongo Db and java		
10	Develop a CRUD Application using Mongo Db(Mini Project)		

Ex. Nos. 1	MONGO DB INSTALLATION AND BASIC SET UP
Date :	

AIM

To install Mongo DB and do the basic set up

PROCEDURE

1. System Requirements and Pre-installation Checklist

Supported Windows Versions: Windows Server 2022, Windows Server 2019, Windows 11 (64-bit, x86_64).

WSL Not Supported: MongoDB does not support Windows Subsystem for Linux (WSL).

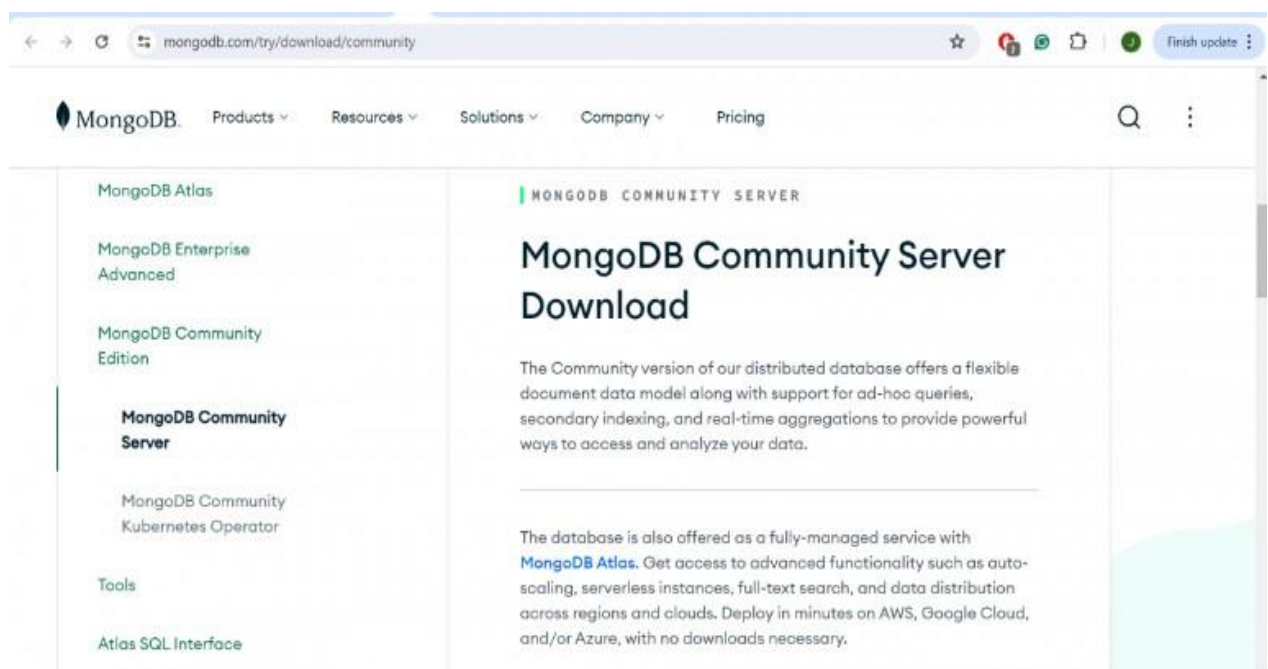
Disk Space: Minimum 10 GB of free space (additional space for data).

Hardware Requirements:

- RAM: Minimum 4 GB.
- CPU: Intel (Sandy Bridge or later), AMD (Bulldozer or later).

2. Downloading and Setting Up MongoDB

Go to the MongoDB official download page to access the MongoDB Community Server section and begin downloading the installation package for Windows.

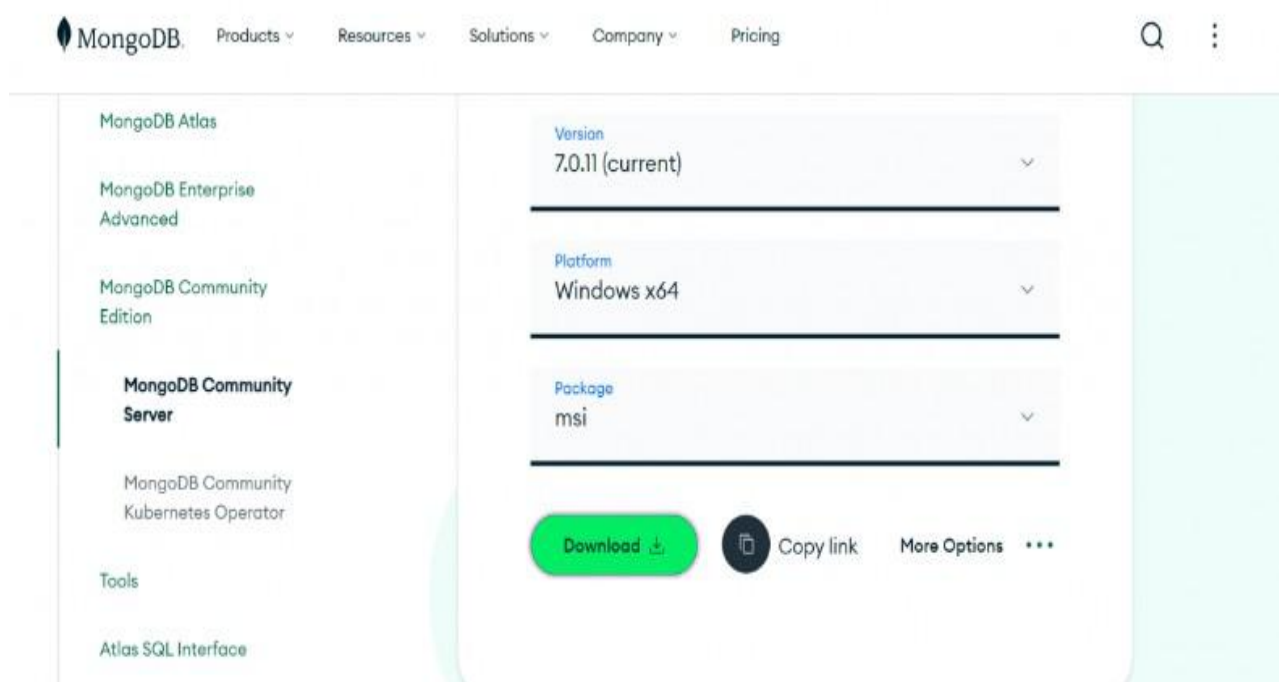


3. Selecting the Correct MongoDB Version and downloading the MongoDB Installer (MSI Package)

- ❖ Choose the latest stable release for optimal performance and security, unless you need an older version.
- ❖ Ensure you select the 64-bit version of MongoDB, as 32-bit versions are not compatible with MongoDB.

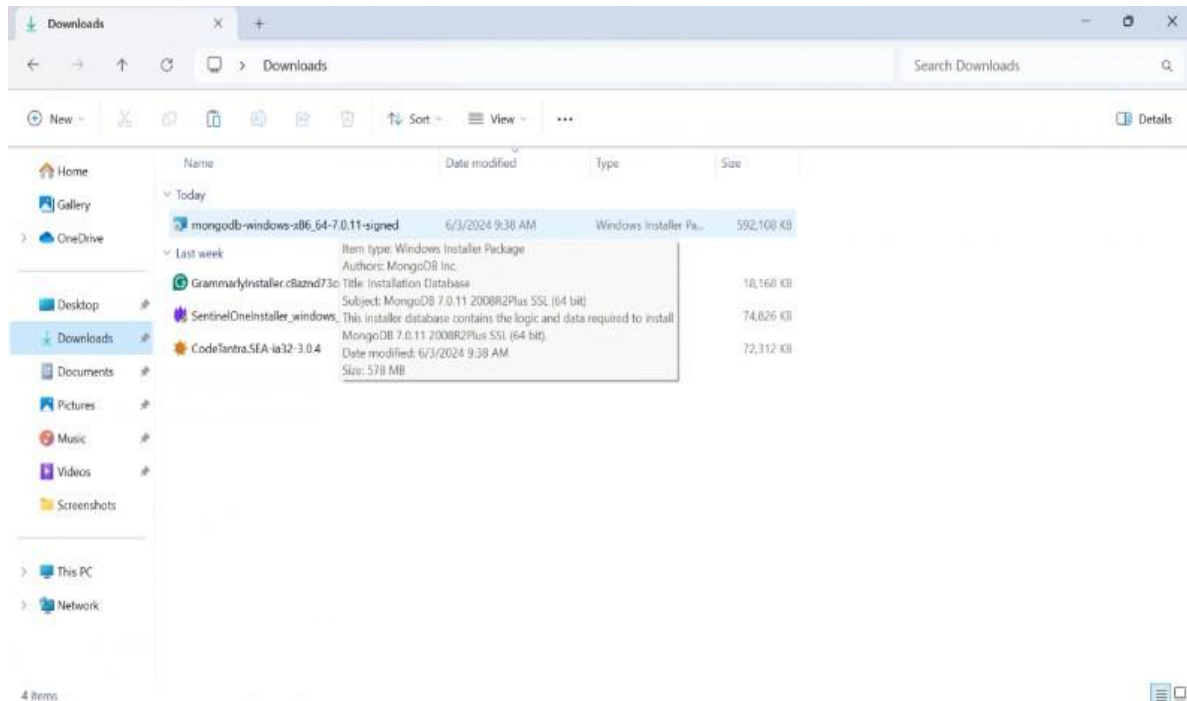
Downloading:

- ❖ Select the MSI package type for Windows and click Download.
- ❖ The .msi installer file will download to your default location (usually the Downloads folder).
- ❖ Double-click the .msi file to launch the installation wizard and begin setup.

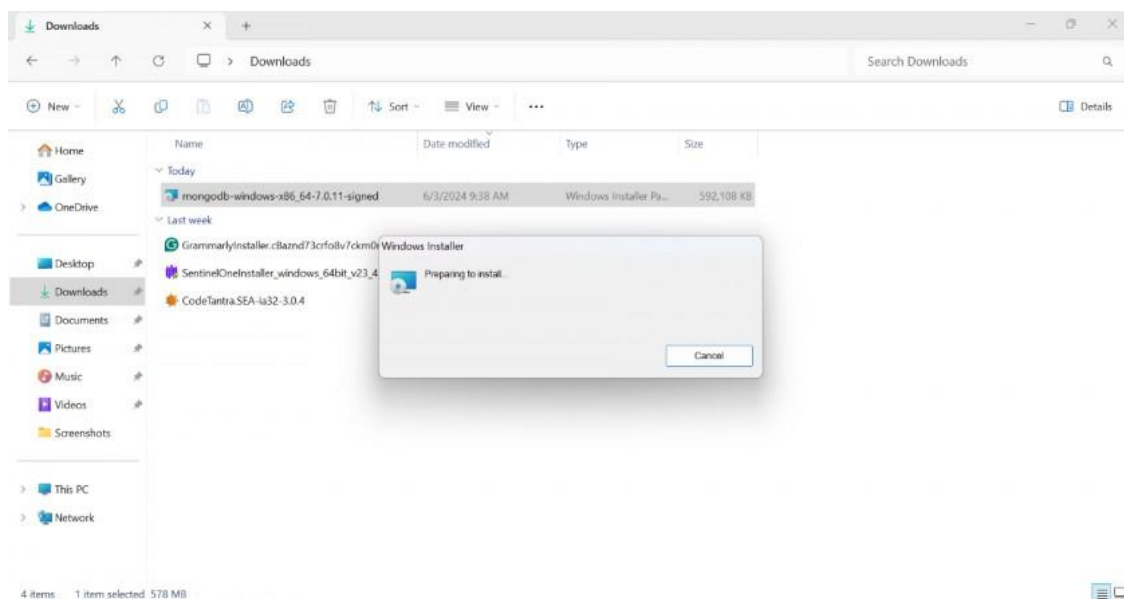


4. Installing MongoDB on Windows

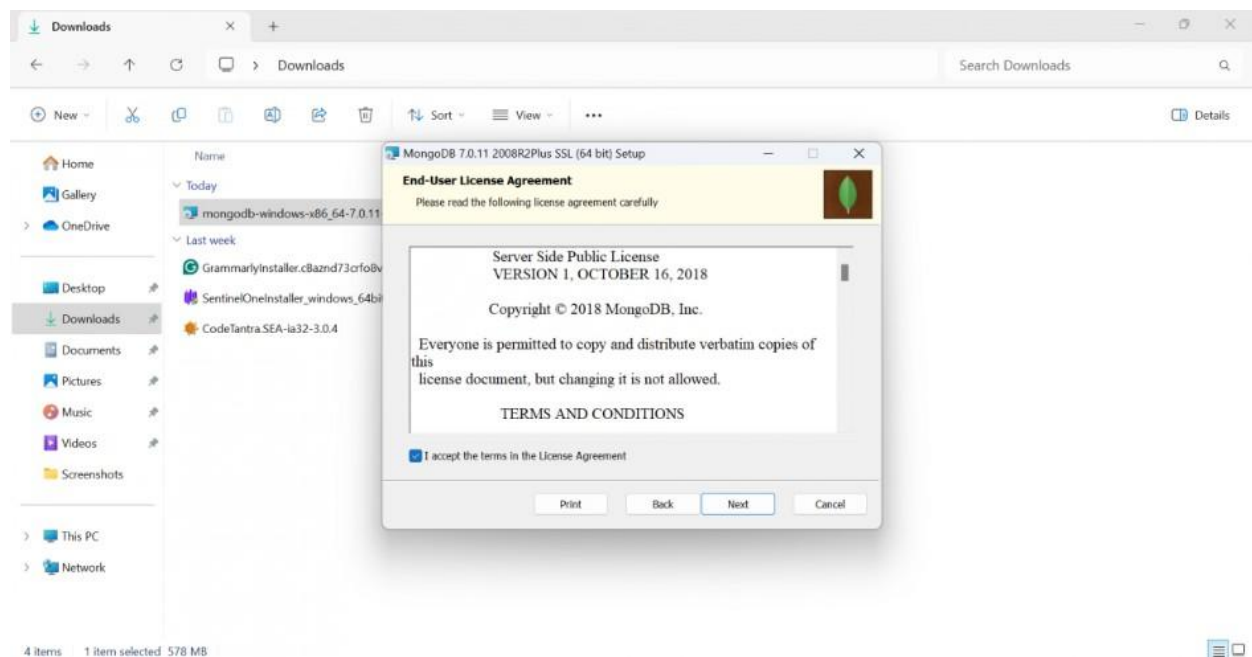
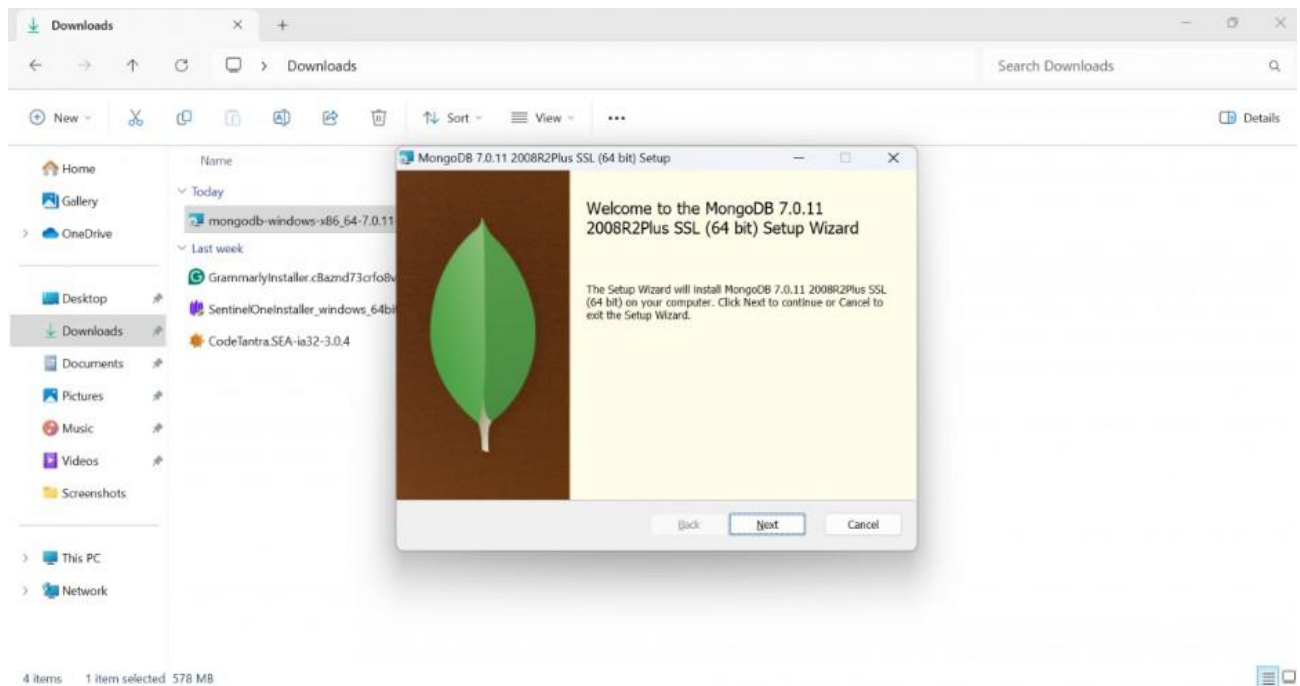
- i. Locate the downloaded .msi file in the Downloads folder.



- ii. Double-click the file to start the installation wizard.

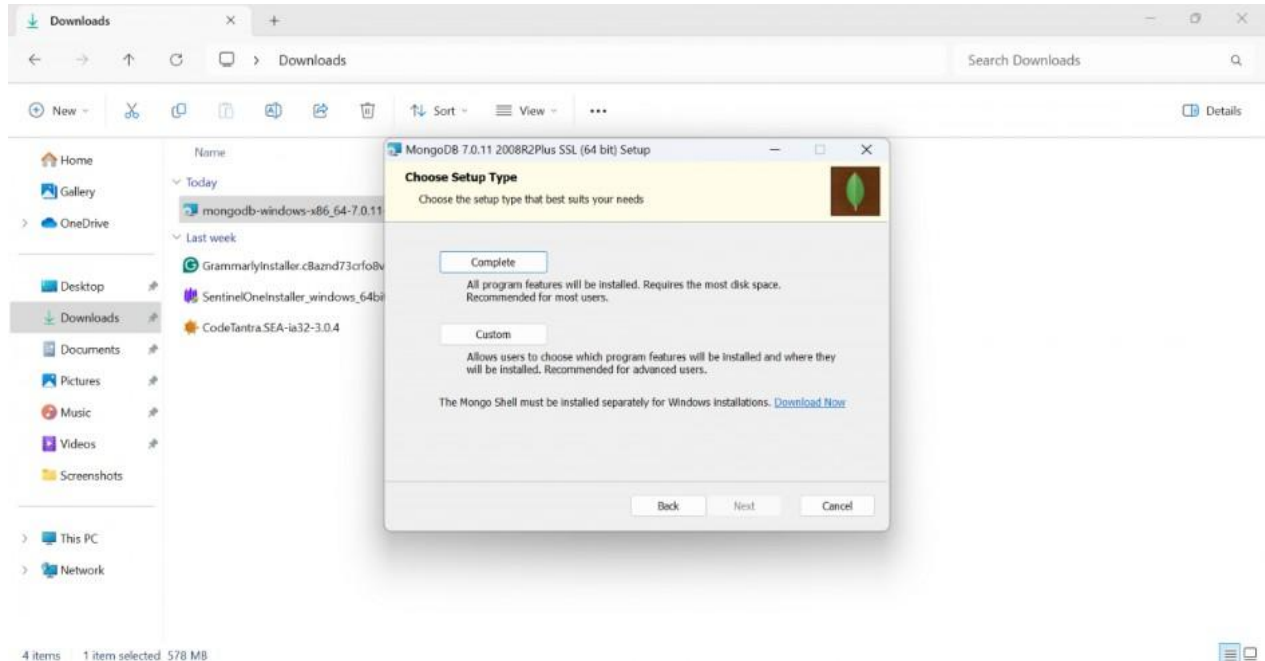


iii. Follow the prompts in the wizard to complete the setup.



5. Choosing the Installation Type

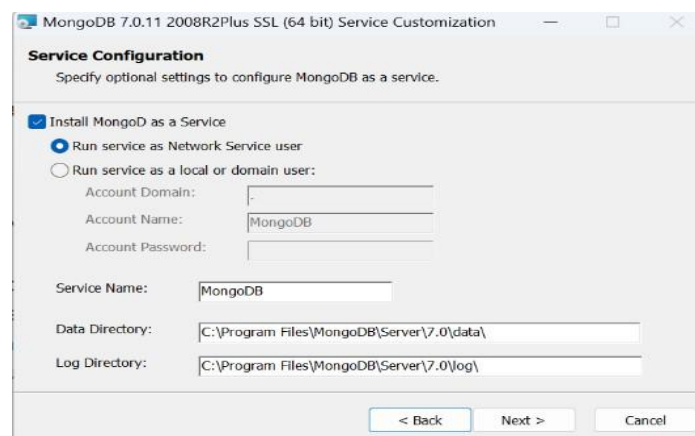
- i. Choose Complete Setup for a default installation with all features (recommended).
- ii. Select Custom Setup to choose specific components and installation locations.



6. Configuring Network Options and Data Directories:

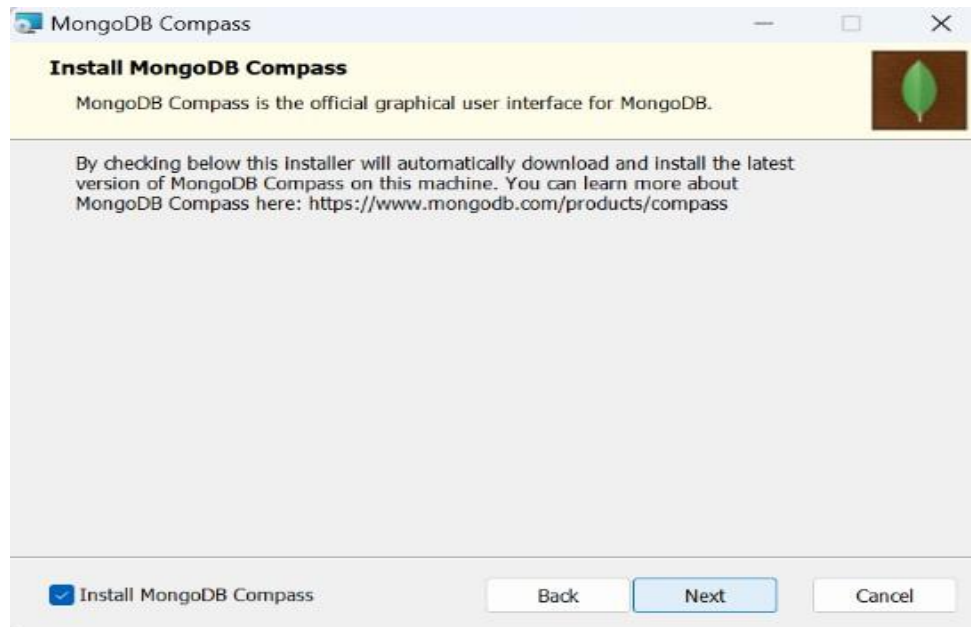
Service & Network Setup: Choose to run MongoDB as a service and configure network settings (localhost or custom IP).

Install & Launch: Complete installation, then start MongoDB with mongod and verify it's running with the "waiting for connections" message.

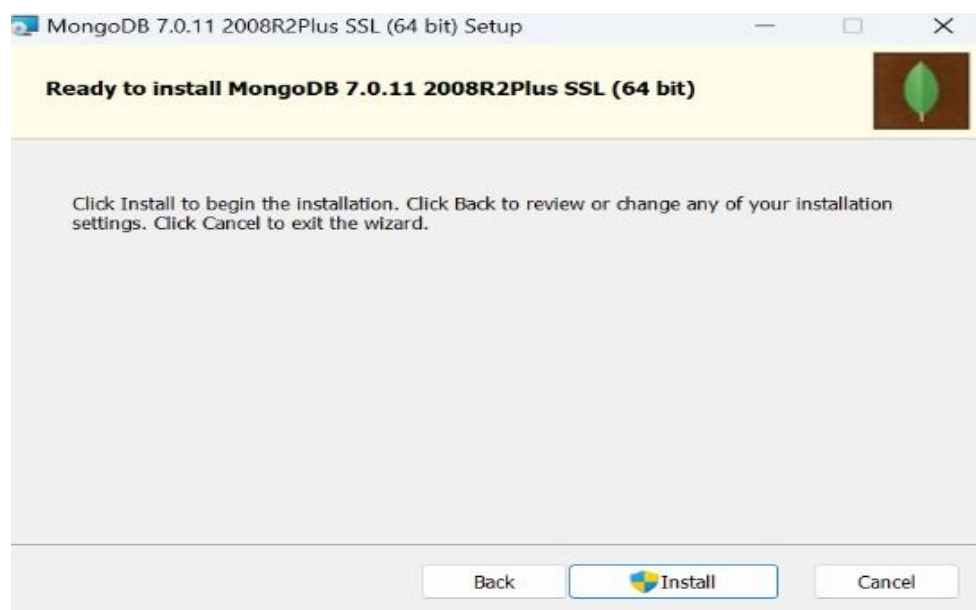


7. Completing the Installation Process

- i. **Install MongoDB Compass (Optional):** You can choose to install MongoDB Compass along with the MongoDB server. This GUI tool helps in managing your database.



- ii. **Finalize Installation:** Review your settings and click 'Install' to begin the installation process. A progress bar will display the installation status.



- iii. **Launch MongoDB:** Once the installation is complete, you can start MongoDB using the `mongod` command in the command prompt if it's not set to run as a service. Ensure that the `mongod.exe` process is running successfully by checking for the 'waiting for connections' message in the console output.



MongoDB Compass is being installed.

It will launch once it is done.

iv. **Mongo DB Start Screen:**

